

K12 Audit Response Re-Audit: LPS Backend Policy and Auto-Dimension Contract

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Verdict

Partially accepted. The two K12-specific blockers from the prior audit are resolved:

1. `local.pca.real.data.contract` now excludes `local.chart.method = "second.order.svd"`.
2. The accidental LPS native diagnostic `.Rd` files are no longer generated, and the K12 NAMESPACE change for `lps.backend.diagnostics` is clean.

However, the audit response is not ready to accept as a package handoff because it substantially expanded scope beyond K12. It moves the `gflow` parity tests out of the ordinary suite, removes `gflow` from `Suggests`, changes graph-geodesic bridging to `dgraphs`, adds a new harmonic smoother native backend, and introduces a new generated internal Rcpp man page. Those changes may be the right direction, but they need their own handoff/audit contract rather than being bundled into the K12 response.

Findings

P1: Response hides parity skips from `make test` without adding an explicit `migration-test` target

The ordinary `make test` target runs only `tests/testthat` (`/Users/pgajer/current_projects/geosmooth/M`). The response moves the GE3/GE4 `gflow` parity tests from `tests/testthat` into `tests/migration`, so the ordinary suite now reports zero skips. Running the migration suite directly still yields the same nine `gflow`-reference skips:

```
[ FAIL 0 | WARN 0 | SKIP 9 | PASS 13 ]
```

This is not a K12 backend-policy blocker, but it is a package-audit blocker for the response as submitted. The response should either:

- add an explicit Makefile target such as `make test-migration` and document that parity tests are opt-in migration checks; or
- leave the parity tests in the ordinary suite until a separate `migration-test` policy is approved.

Otherwise, “ordinary suite has zero skips” is achieved by moving parity coverage out of the checked path, not by resolving the parity preconditions.

P1: Response includes broad non-K12 graph and harmonic backend changes

The response modifies code well beyond K12 diagnostics:

- DESCRIPTION removes `gflow` from `Suggests`.
- `R/split_bridge_helpers.R` replaces `gflow` bridge helpers with package-local graph payload validation and `dgraphs::shortest.path`.
- `R/malps.R`, `R/lpl_tf.R`, and `R/ssrhe_hessian_energy.R` documentation now describe `dgraphs` ownership.
- `R/harmonic_smoother.R` switches from a `gflow` native symbol call to a new package-local `rcpp_harmonic_smoother`.
- `src/harmonic_smoothing_native.cpp` adds a substantial topology-tracking backend implementation.
- graph-trend-filtering and graph-boundary tests are rewritten around the new package/`dgraphs` boundary.

These changes may be necessary for the broader `geosmooth` split, and the tests I ran did pass. But they are not just an audit response to the K12 LPS backend policy. They should be split into a separate graph-boundary/native-harmonic handoff with parity expectations, especially because `AGENTS.md` says not to silently change smoother semantics during migration.

P2: Generated internal native documentation drift persists for a new wrapper

The previous audit flagged accidental generated `.Rd` files for internal LPS native diagnostic wrappers. Those two files are gone after `make document`. But the response introduces another internal Rcpp wrapper, `rcpp_harmonic_smoother()`, and `make document` generates:

`man/rcpp_harmonic_smoother.Rd`

The page says it is an internal C++ backend for `harmonic.smoother()`. If this is intended as an internal native wrapper like the LPS diagnostic wrappers, mark it `@noRd`; if it is intentionally documented, list it in the handoff and explain why this internal wrapper should have an `Rd` page.

Resolved K12 Items

- The contract predicate now requires ordinary PCA charting: `local.chart.method.effective == "pca"` in addition to `local-PCA`, `chart.dim = "auto"`, support metric `both`, and selection metric `operator`.
- The GE7 regression test now verifies that `local.chart.method = "second.order.svd"` has `chart.dim.auto = TRUE` but `local.pca.real.data.contract = FALSE`.
- Direct audit probe confirmed:

```
local.chart.method.effective = second.order.svd
chart.dim.auto = TRUE
local.pca.real.data.contract = FALSE
```

- The earlier accidental generated files `man/rcpp_kernel_local_polynomial_cv_local_pca_profile.Rd` and `man/rcpp_kernel_local_polynomial_neighbor_probe.Rd` no longer appear after `make document`.

Verification

Passed:

`make document`

```
Rscript -e 'pkgload::load_all(".", quiet = TRUE); testthat::test_file("tests/testthat/test-g
```

```
Rscript -e 'pkgload::load_all(".", quiet = TRUE); testthat::test_file("tests/testthat/test-g
```

```
Rscript -e 'pkgload::load_all(".", quiet = TRUE); testthat::test_dir("tests/testthat")'
```

```
Rscript -e 'pkgload::load_all(".", quiet = TRUE); testthat::test_dir("tests/migration")'
```

```
R CMD build .
```

```
R CMD check geosmooth_0.0.0.9000.tar.gz --as-cran --no-examples --no-tests --no-manual
```

```
git diff --check
```

Results:

- Focused GE7 test: passed.
- Focused graph-trend-filtering test: passed.
- Ordinary `tests/testthat`: 0 failures, 0 warnings, 0 skips, 916 passes.
- Migration tests: 0 failures, 0 warnings, 9 skips, 13 passes.
- R CMD `check --no-tests`: completed with 1 WARNING, 1 NOTE. The warning is the existing development/non-mainstream dependency warning; the note is the existing non-standard top-level `scripts` directory.

Recommendation

Accept the narrow K12 LPS diagnostic fix after adding/confirming the generated files intended for K12. Do not accept the full response bundle as a single K12 handoff yet. Split the graph-boundary, `dgraphs`, migration-test-policy, and native harmonic smoother changes into a separate handoff with its own parity and API contract, and either add a `test-migration` target or explicitly document that migration parity is no longer part of ordinary package QA.